

Optimizing Delivery Cost and Time Using Real-World Data: A Comparison of Different Optimization Methods

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Abstract

This project focuses on solving a multi-objective optimization problem using real-world logistics data. The objectives include minimizing cost, delivery time, and emission using Scalarization, NSGA-II, and PSO.

1 Introduction

Multi-objective optimization is widely used in logistics and decision-making problems. This project minimizes cost, time, and emission simultaneously.

2 Dataset Description

The dataset is derived from the Brazilian e-commerce dataset :contentReference[oaicite:0]index=0, containing order, customer, and delivery information. A subset of 200 samples with features such as cost, time, distance, weight, and zone is used for optimization.

2.1 Features

- Cost , Delivery Time , Distance , Weight , Zone

3 Mathematical Model

Decision variable:

$$x_i \in \{0, 1\}$$

Objective:

$$\min \sum (w_1 c_i + w_2 t_i + w_3 e_i) x_i$$

Constraints:

$$\sum x_i \leq 50$$
$$\sum w_i x_i \leq 300$$

4 Methodology

4.1 Scalarization

A weighted sum approach is used to convert the multi-objective problem into a single objective optimization problem under given constraints.

4.2 Greedy

A baseline heuristic that selects items with minimum cost while satisfying capacity and selection constraints.

4.3 NSGA-II

A multi-objective evolutionary algorithm that generates a diverse set of Pareto-optimal solutions using non-dominated sorting.

4.4 PSO

A population-based optimization technique where particles iteratively update positions to minimize a scalarized objective function.

4.5 Convergence Plot

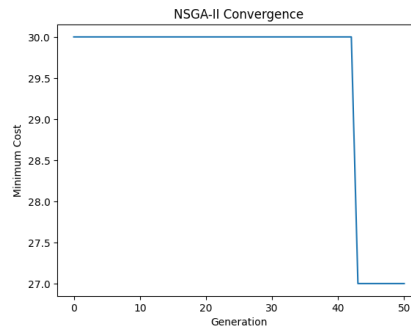


Figure 1: NSGA-II Convergence Plot showing minimum cost vs generations

4.6 Pareto Front (Cost vs Time)

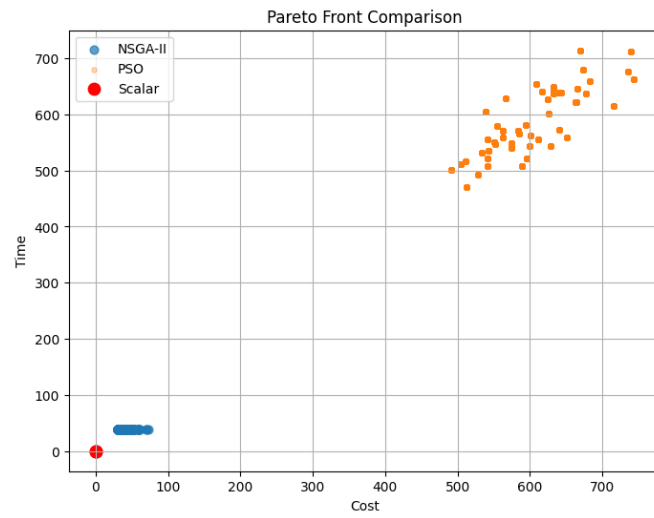


Figure 2: Pareto front comparison between NSGA-II, PSO, and Scalarization

4.7 3D Pareto Front

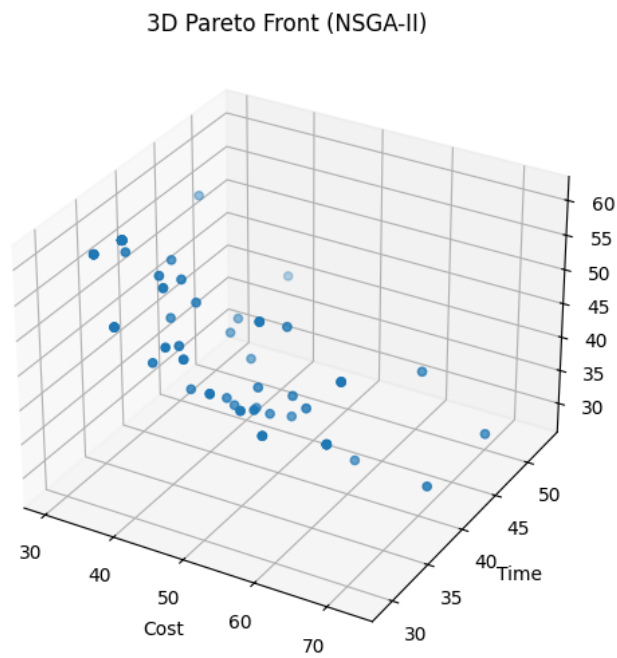


Figure 3: 3D Pareto front showing Cost, Time, and Emission (NSGA-II)

5 Results

5.1 Final Comparison Table

Method	Cost	Time	Emission
Scalarization	0.0	0.0	0.0
Greedy	442.71	402.0	24669.77
NSGA-II	30.0	29.0	28.0
PSO	491.0	501.0	509.0

Table 1: Final Performance Comparison

6 Conclusion

NSGA-II provides better Pareto solutions compared to deterministic methods.

7 Future Work

- Larger dataset
- Deep learning integration
- Hybrid optimization

References

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